

Greyscale gambit

Answer: 8/8/2p1r3/1pPK4/N7/p1B5/k7/1b2r1b1

Solution: First assume, for the sake of contradiction, that all the statements are true. Then, in particular, statement 4 implies that if two pieces lie on the same diagonal, they must be the same colour. Therefore, no bishop can threaten a king, making the statement 11 false, which is a contradiction. Therefore all of the statements are false.

We will now determine which pieces must be of the same colour. To represent this, we will mark some pieces blue and others green, indicating which ones share a colour. Later, we will deduce whether green corresponds to white or black, thereby reconstructing the chess game setup. The negation of statement 11 implies that if the king and bishop are adjacent diagonally, then they must be the same colour. The negation of statement 7 implies that bishops on squares of the same colour are of different colours. From this, we can deduce the colours of the remaining bishops (see Figure 1). The negation of statement 9 implies that the knight on the bottom row must be green. Consequently, the negation of statement 8 implies that the knight in the third column must be blue (see Figure 2).

Figure 1

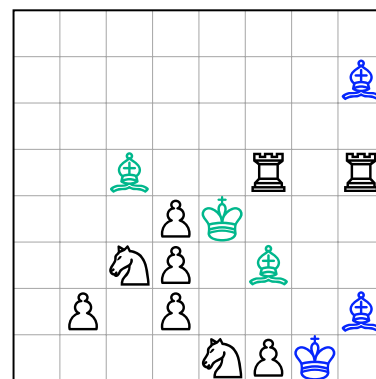
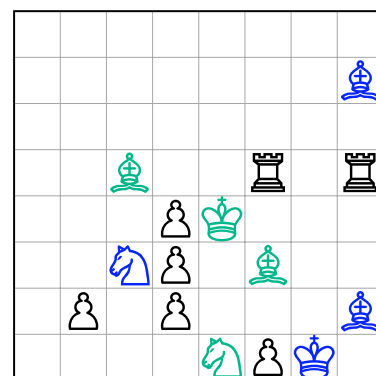


Figure 2



The negation of statement 6 implies that the two pawns at the top of the fourth column are green, and the pawn in the sixth column is blue (see Figure 3). The negation of statement 5 implies that the remaining pawn in the fourth column must be blue – otherwise, the knight would not be threatened by a pawn (see Figure 4).

Next, the negation of statement 7 implies that there have been no promotions. Thus, the only way a pawn could have reached the bottom row is if the board had been rotated by 90 degrees. Recalling from the negation of statement 5 that a pawn is threatening the green knight, we conclude that the blue pieces are moving from west to east and the green pieces from east to west.

The negation of statement 10 implies that there is at least one piece that has not moved. The only possible candidate is the remaining pawn, which must be blue to satisfy this condition (see Figure 5).

Finally, the negation of statement 3 implies that the remaining two rooks are of the same colour, and the negation of statement 2 then implies that the rooks must be blue. This completes the full board setup (see Figure 6).

Figure 3

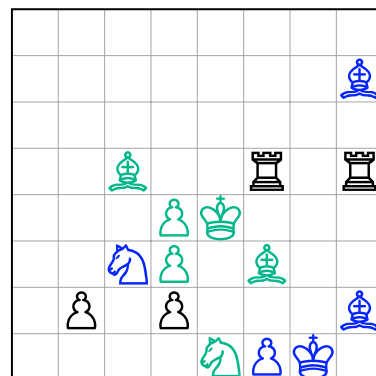


Figure 4

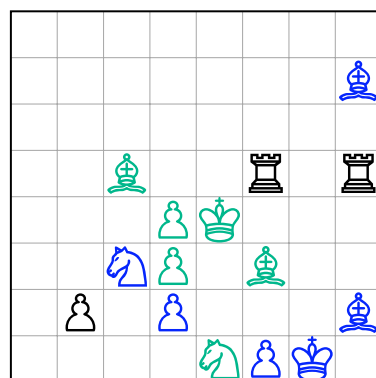


Figure 5

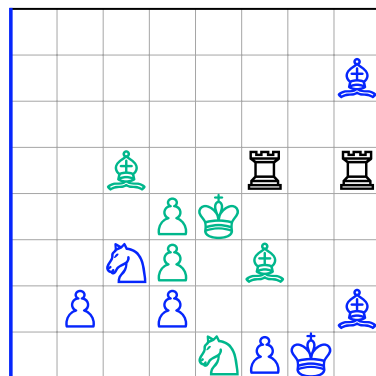
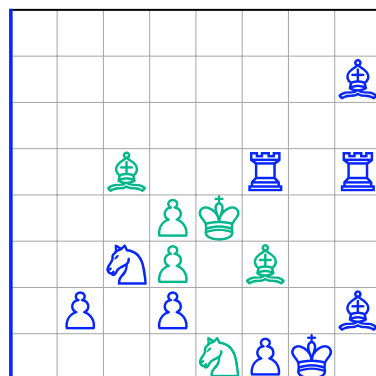


Figure 6



Now note that the green king is in check, so it must be green's turn. However, the negation of statement 1 implies that it is White's turn. Therefore, we conclude that green represents White and blue represents Black. This completes the full board setup with the correct colours (see Figure 7).

White is in check from the black knight. The king has no legal moves, so the only way out is to capture the knight with the pawn (see Figure 8).

Now Black is in check from the white bishop.

This check cannot be blocked by any of the black pieces, and the black king cannot move, so the only way out is to capture the bishop with the black rook (see Figure 9).

Next, the white king is in check from the black bishop. This check also cannot be blocked by any of the white pieces, so the only option is to move the king, with D5 being the only available square (see Figure 10).

At this point, Black can deliver checkmate by capturing the white pawn with the black pawn. A quick analysis confirms that this is the only way to achieve checkmate in this move, thus this is precisely the move that would have been played. (see Figure 11).

This chessboard setup translates to 8/8/2p1r3/1pPK4/N7/p1B5/k7/1b2r1b1 in the chess FEN notation.

Figure 7

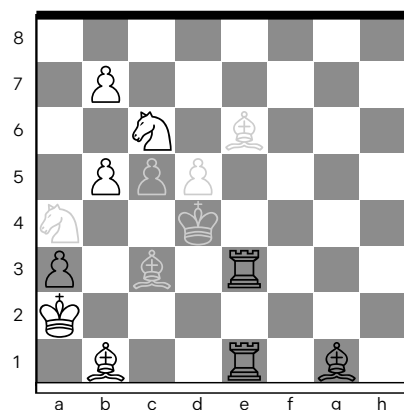


Figure 8

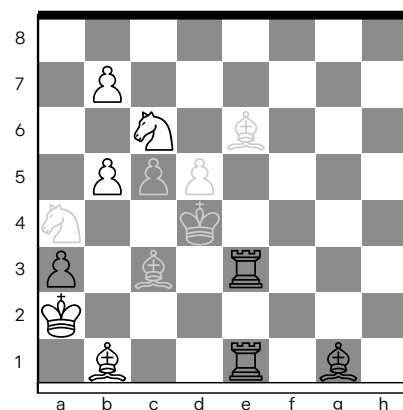


Figure 9

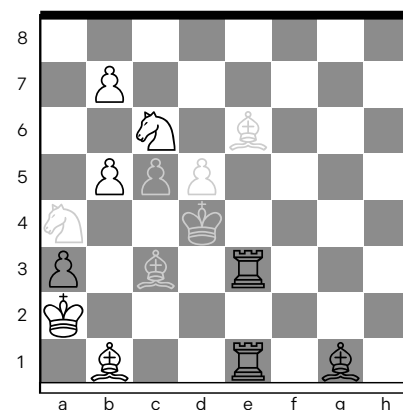


Figure 10

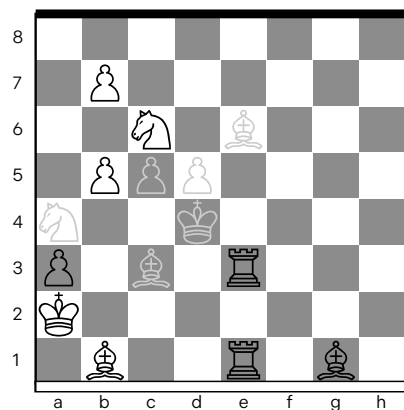


Figure 11

